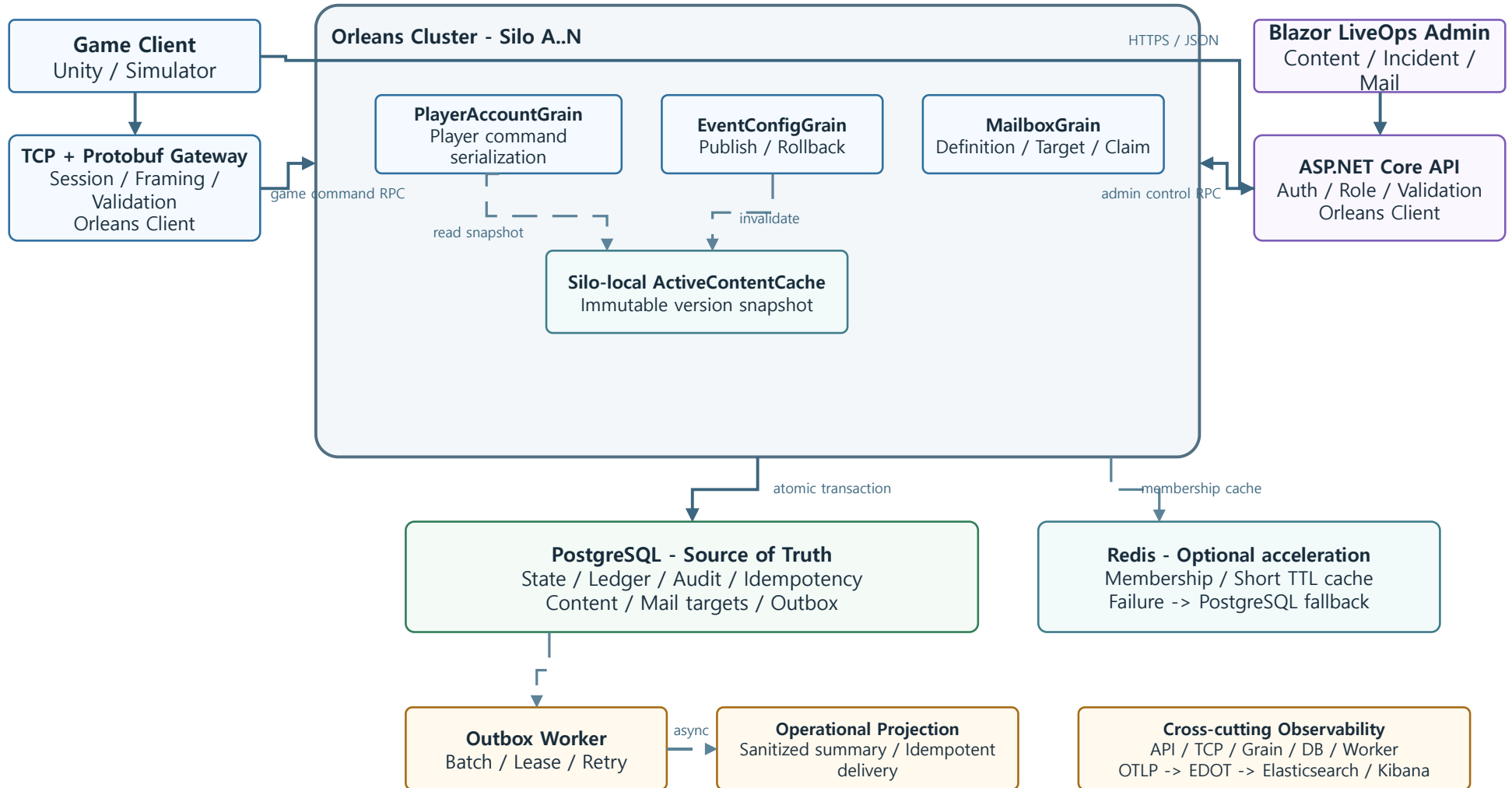


8.1 전체 구성

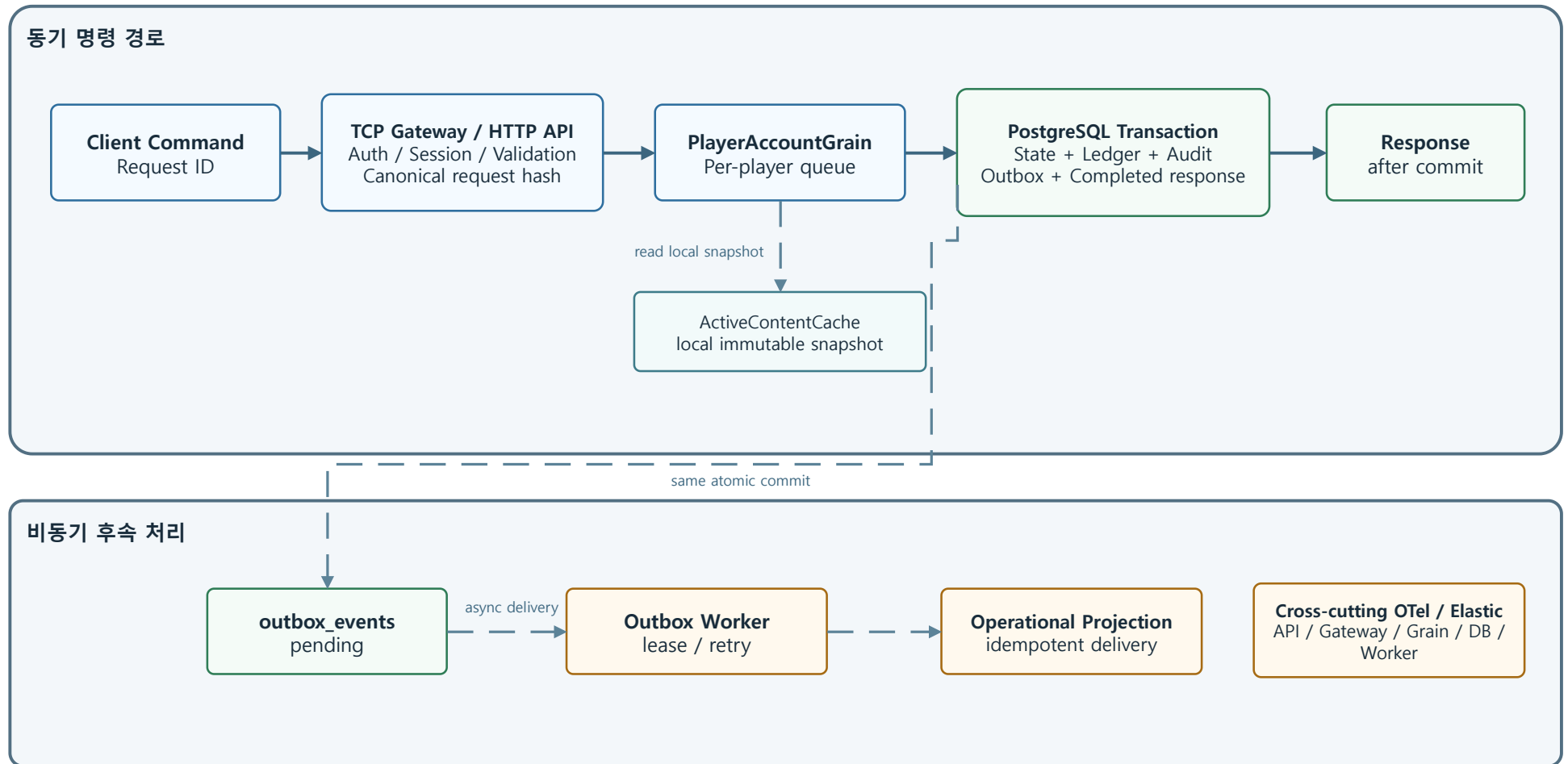
게임 명령 경로와 운영 제어 경로는 분리하고, PostgreSQL commit 이후 비동기 처리를 분기한다.



Solid: command / transaction | Dashed: cache / async | Observability is cross-cutting, not an Outbox destination

8.2 동기 요청과 비동기 분기

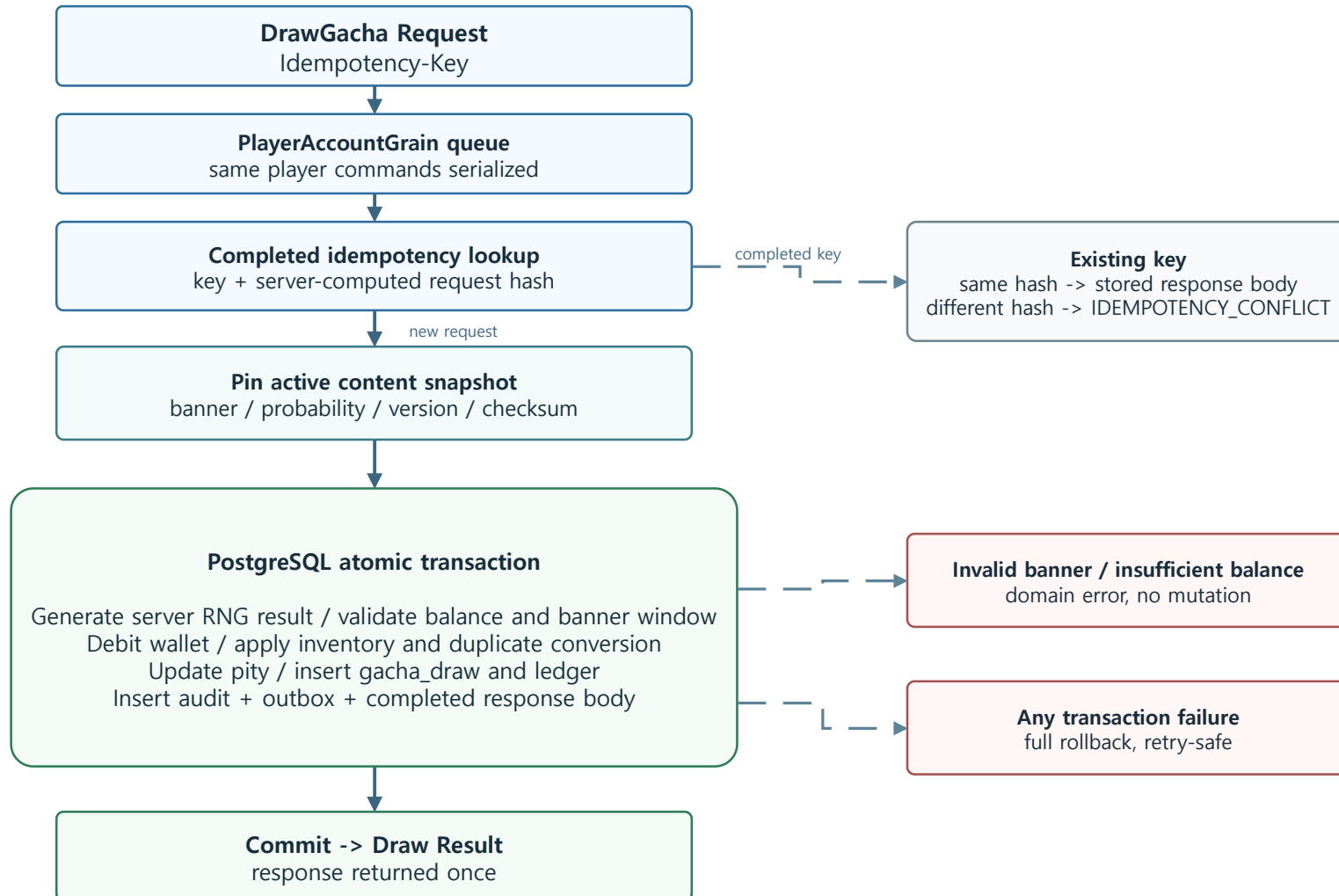
Worker와 Observability를 동기 요청의 직렬 단계로 오해하지 않도록 실행 경계를 분리한다.



The synchronous response path ends at PostgreSQL commit; asynchronous delivery failures do not roll back player state.

8.3 가차 원자 트랜잭션

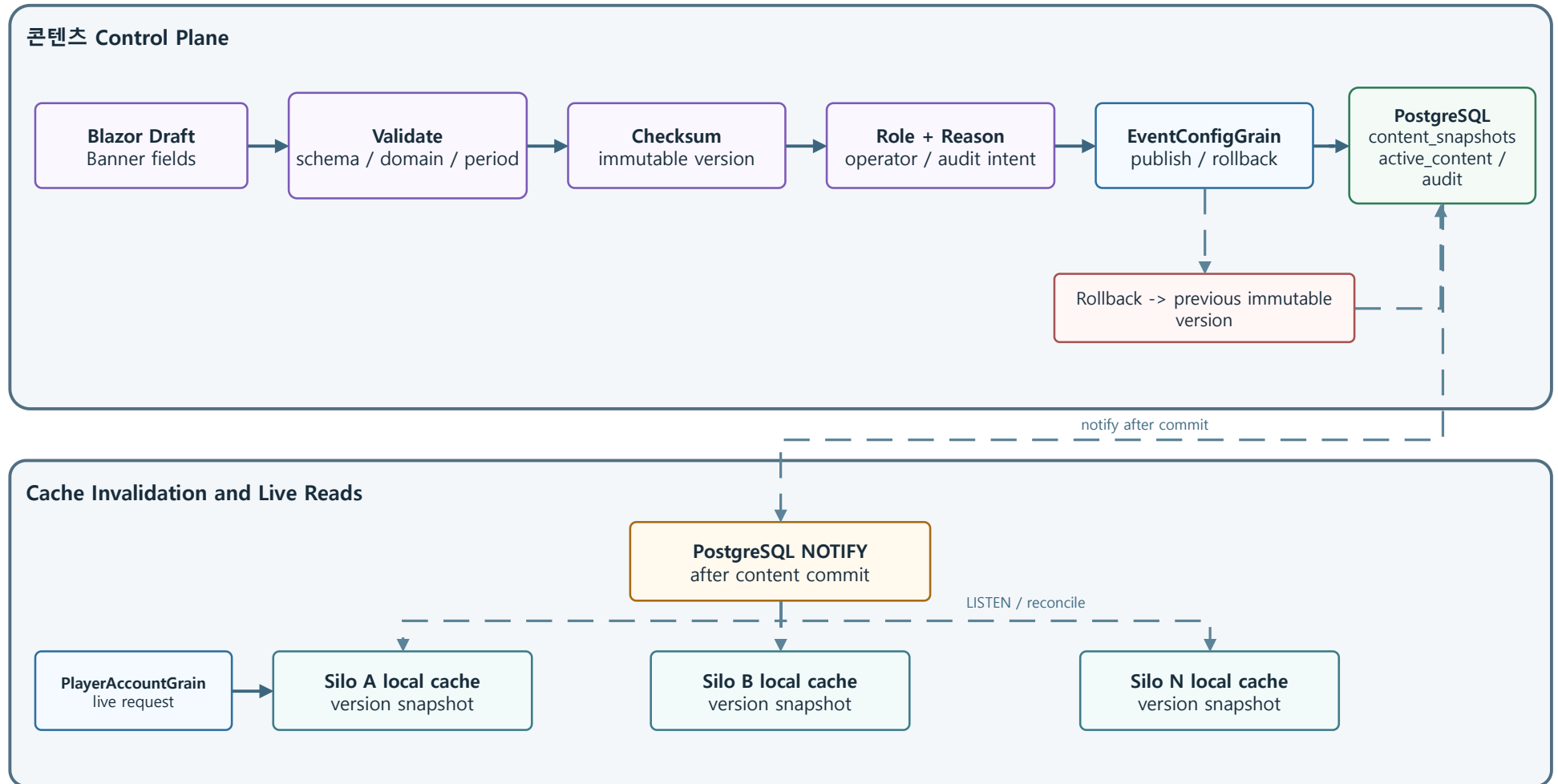
가차 결과, 재화, 인벤토리, 천장, 이력, Outbox, 먹등성 응답을 하나의 commit으로 묶는다.



Completed replay returns before reading current content; request idempotency stores no PENDING lock.

8.4 콘텐츠 배포와 Silo 캐시

EventConfigGrain은 control plane에만 두고, live request는 각 Silo의 local snapshot을 읽는다.

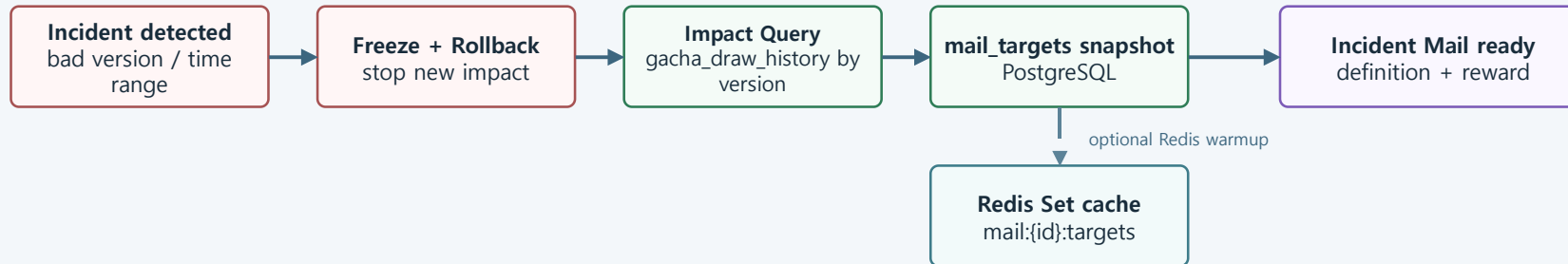


Hot path: PlayerAccountGrain -> local cache. It does not call EventConfigGrain for every request.

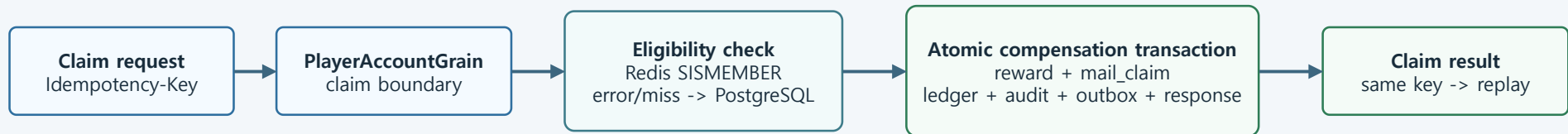
8.5 사고 복구와 보상

잘못된 콘텐츠를 되돌리는 것과 이미 영향을 받은 플레이어들을 보상하는 것을 별도 단계로 처리한다.

운영 사고 대응



플레이어 보상 수령

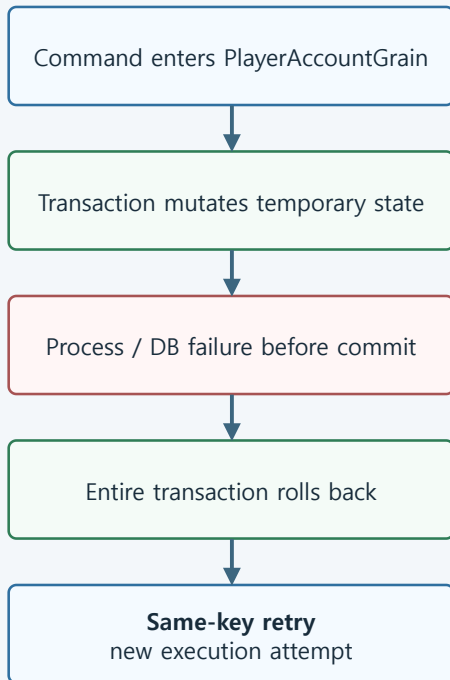


Redis is never the compensation source of truth; connection failure must fall back to PostgreSQL.

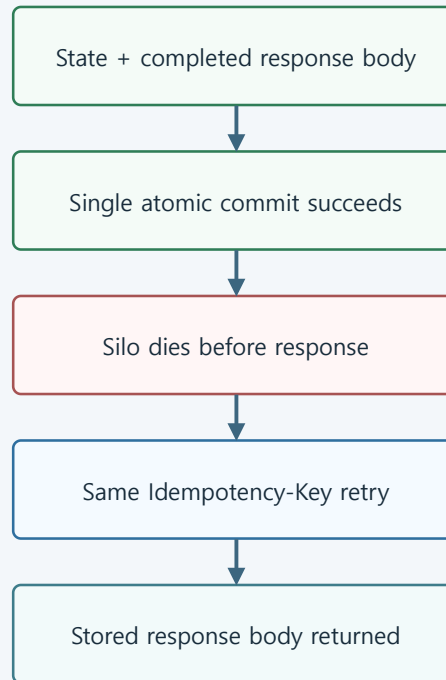
8.6 멍등성과 장애 복구

장애 시점을 분리해 rollback, response replay, Outbox retry의 책임을 명확히 한다.

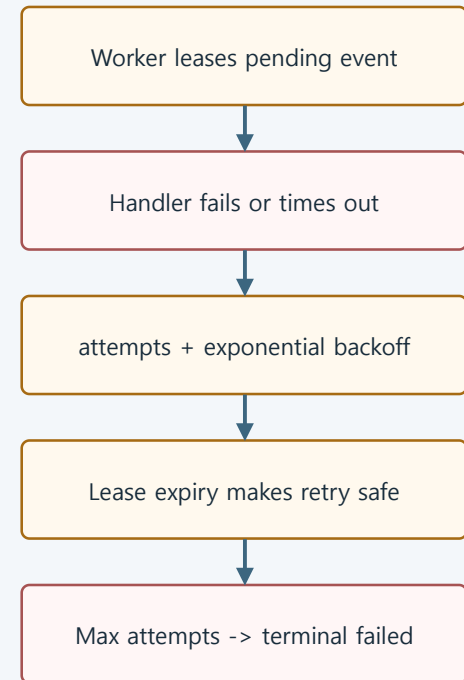
A. Commit 전 장애



B. Commit 후 응답 전 장애



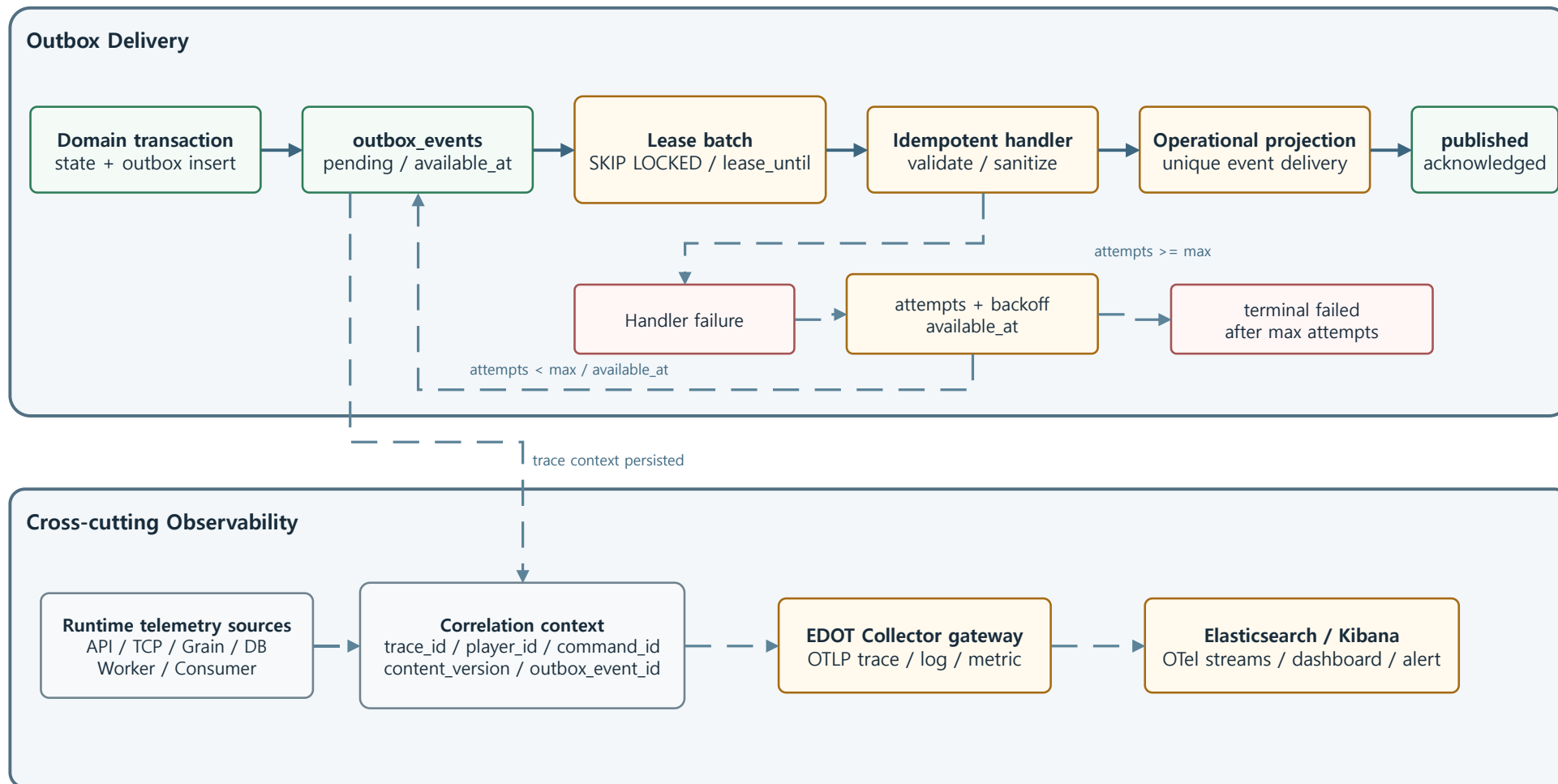
C. Outbox 처리 장애



Request idempotency: COMPLETED only, no PENDING | Outbox processing state: pending / processing / published / failed

8.7 Outbox와 관측성

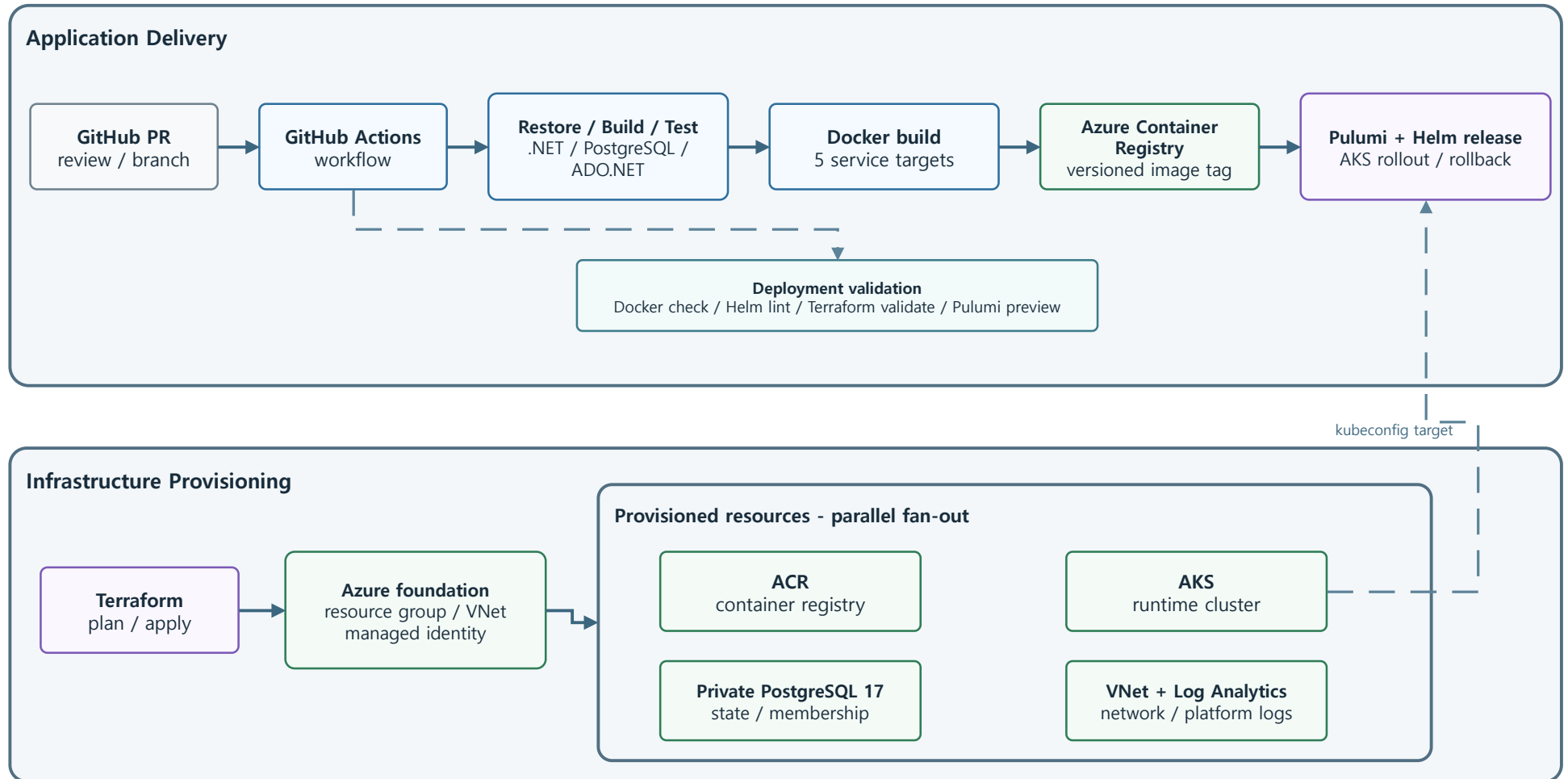
Outbox는 idempotent consumer로 전달하고, Observability는 모든 실행 단계를 횡단해 관찰한다.



Persist trace context with the Outbox event so Worker spans can be linked to the originating command.

8.8 CI/CD와 Azure 배포

런타임 요청 흐름과 배포/IaC 흐름을 분리해 배포 책임과 검증 경계를 명확하게 보여준다.

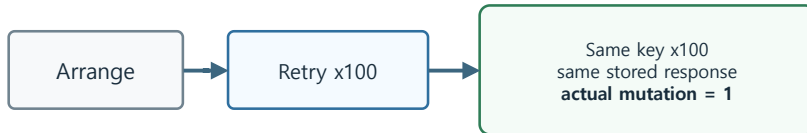


Local: Docker Compose | CI: GitHub Actions | Registry: ACR | Runtime: AKS | IaC: Terraform | App release: Pulumi + Helm

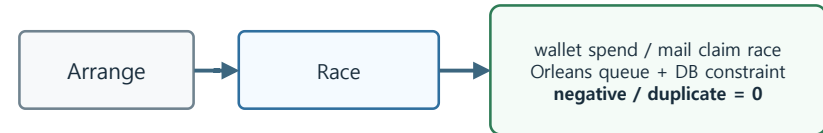
8.9 핵심 검증 시나리오

아키텍처 주장을 기능 개수 대신 실패·동시성·복구 테스트로 증명한다.

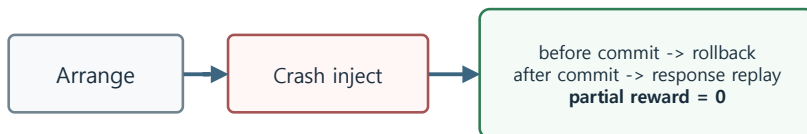
1. Idempotency Replay



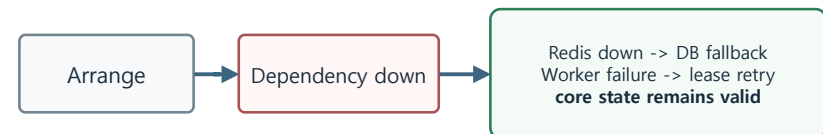
2. Concurrent Player Commands



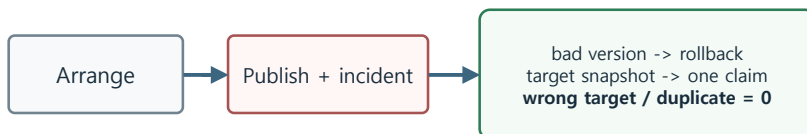
3. Crash Windows



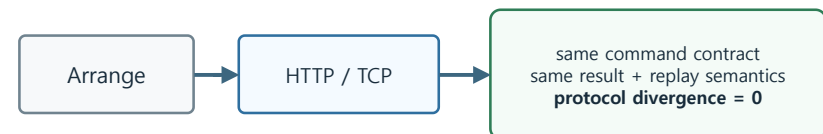
4. Dependency Failure



5. Content Rollback + Compensation



6. HTTP / TCP Command Parity



Verification evidence - xUnit 91 / PostgreSQL 17 / ADO.NET 2-Silo + E2E 4 / Docker 5 targets / Helm lint / Terraform validate / Pulumi preview

Tests must fail visibly when dependencies are not available; gated tests must not silently return as passed.